



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Make-Up Test (Session: 2025-26)

Roll No.:

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Course Name: B.Tech
 Subject Name:
 Subject Code: 25AS102

Max. Marks: 15
 Time: 75 Minutes
 Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 6)				
(a)	For what value of a the vectors $(1, 1)$ and $(1, a)$ in $V_2(F)$ are linearly dependent.	CO1	K2	2.1.3
(b)	For what value of k will the vector $(1, -2, K)$ in $V_3(\mathbb{R})$ be the linear combination of vectors $(3, 0, -2), (2, -1, -5)$.	CO1	K2	2.1.3
(c)	Find the nullity of linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x_1, x_2) = (x_1 + x_2, x_1)$.	CO2	K2	2.4.1
(d)	Define Basis of a vector space.	CO2	K1	2.4.1
(e)	Find the eigen values of the matrix $\begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$.	CO3	K2	2.1.3
(f)	If iA is a skew Hermitian matrix, prove that A is a Hermitian matrix.	CO3	K2	2.1.3
SECTION-B				
2. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Show that the set $B = \{(2, 1, 4), (1, -1, 2), (3, 1, -2)\}$ form a basis of vector space $\mathbb{R}^3(\mathbb{R})$.	CO1	K3	2.1.3
(b)	Determine a basis and dimension of the sub space W spanned by the vectors $(-3, 1, 2), (0, 1, 3), (2, 1, 0), (1, 1, 1)$.	CO1	K3	2.4.1, 2.4.2
SECTION-C				
3. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Show that the mapping $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (2x - 3y, 7y + 2z)$ is linear transformation.	CO2	K2	2.4.1, 2.4.4
(b)	Consider the following two basis of $\mathbb{R}^2$: $S = \{u_1, u_2\} = \{(1, 2), (3, 5)\}$ and $S' = \{v_1, v_2\} = \{(1, -1), (1, -2)\}$. Find the change of basis matrix P from S to S' .	CO2	K3	2.4.2

SECTION-D				
4. Attempt Any one Question			(Same K Levels) (3 x 1)	
(a)	Find the rank of matrix $\begin{bmatrix} 2 & -1 & 3 & 1 \\ 1 & 4 & -2 & 1 \\ 5 & 2 & 4 & 3 \end{bmatrix}$	CO3	K2	2.4.1, 2.4.4
(b)	Solve the system of equations using matrix method $\begin{aligned} x_1 + 2x_2 + x_3 &= 2, & 3x_1 + x_2 - 2x_3 &= 1, \\ 4x_1 - 3x_2 - x_3 &= 3, & 2x_1 + 4x_2 + 2x_3 &= 4. \end{aligned}$	CO3	K3	2.4.2

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create